Prodigy Anesthesia Quick Review

Central Nervous System Physiology

What is the normal cerebrospinal fluid pressure?	When lying in a horizontal position, the normal cerebrospinal fluid pressure is about 130 mm of water or 10 mmHg. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 747.*
What are some common causes of increased cerebrospinal fluid pressure?	A brain tumor can result in the decreased reabsorption of the CSF, which can result in an increased CSF pressure. Hemorrhage or infection are other potential causes as the appearance of red blood cells or white cells can obstruct the absorption channels in the arachnoid villi. Hydrocephalus is an inborn cause of increased CSF pressure. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 748.*
Where in the brain are the choroid plexuses located? Which locations are associated with the greatest quantity of cerebrospinal fluid produced?	The choroid plexuses are located in the four ventricles. The ones located in the two lateral ventricles produce the greatest quantity of cerebrospinal fluid. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 746.*
What is the rate of formation of cerebrospinal fluid?	About 500 mL of CSF is formed each day. This translates into about 21 mL/hour. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 746.*
About how much cerebrospinal fluid is in the subarachnoid space at any given time?	About 125 mL *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 746.*
How does secretion of cerebrospinal fluid by the choroid plexus occur?	It is dependent upon the active transport of sodium through the epithelium of the choroid plexus. As the sodium is transported out, chloride is pulled outward as well because of its electrochemical attraction to sodium. The combination of sodium and chloride increases the osmotic pressure on the outside of the plexus which results in the osmosis of water through the plexus membrane. It is the fluid that forms the principal constituent of cerebrospinal fluid. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 747.*
Where is the cerebrospinal fluid secreted and where is it absorbed?	Cerebrospinal fluid is secreted by the choroid plexus and absorbed by the arachnoid villi. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 747.*
What is the brain's preferred energy source?	The brain is able to metabolize some lactate as a source of energy although glucose is the preferred energy source. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 576.*

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How much glucose does the brain utilize each minute?	Brain glucose utilization is about 5 mg/100g/min. This is about 75 mg/min in the average adult. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 576.*
About how long will the glycogen stored inside neurons last if it must be converted to glucose during a period of hypoglycemia?	The neurons only contain about a 2 minute supply of glycogen. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 749.*
What percentage of cardiac output goes to the brain?	The brain only comprises about 2% of body mass but receives about 15% of the cardiac output. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 743.*
What are the four arteries that supply blood to the brain?	The left and right carotid arteries and left and right vertebral arteries. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 743.*
What is the most effective method for protecting the brain during ischemic episodes?	Profound hypothermia is the most effective method for protecting the brain during episodes of ischemia. It is often used for up to 60 minutes of total circulatory arrest for cardiopulmonary or neurosurgical procedures with little evidence of neurologic sequelae. Even mild hypothermia at 33-35 degrees Celsius provides some protective benefits. Although propofol, thiopental, and isoflurane offer many protective advantages, no anesthetic has been shown to consistently provide these effects. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 587.*
What is the normal cerebral blood flow?	The normal blood flow to the brain is about 50-65 milliliters per 100 grams of brain tissue per minute. This amounts to about 750-900 milliliters/minute in the average adult. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 743.*
What is the normal cerebral metabolic rate (CMRO2) in adults?	The average CMRO2 in adults is 3-3.8 mL/100g/min or about 50 mL/min *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 576.*

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How do changes in temperature affect the cerebral metabolic rate?	For every 1 degree Celsius that the core temperature drops, the cerebral metabolic rate decreases 5-7% and vice versa. The EEG becomes isoelectric at a body temperature of 20 degrees Celsius. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 578.*
What is the relationship between the cerebral metabolic rate and glucose consumption by the brain?	CMRO2 maintains a direct relationship with glucose consumption. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 576.*
How is cerebral blood flow calculated?	Cerebral blood flow is calculated as: CBF = Cerebral Perfusion Pressure/Cerebral Vascular Resistance *Mccance KL, Huether SE, Brashers VL, & Rote NS. Pathophysiology: The Biologic Basis of Disease. 6th ed. Philadelphia, PA: Mosby, Inc.; 2010: 557.*
How does hypertension affect the cerebral autoregulation curve?	Normally, the body maintains a constant cerebral blood flow between mean arterial pressures of 60 and 160 mmHg. Chronic hypertension will shift both the upper and lower limits of the cerebral autoregulation curve to the right. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 577.*
How does carbon dioxide affect cerebral blood flow?	CBF is directly proportional to PaCO2 between tensions of 20 and 80 mmHg. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 578.*
Between what blood pressures does the brain maintain cerebral blood flow at a normal level? How does it do this?	Normally, the cerebral blood flow remains constant between mean arterial pressures of 60 to 160 mmHg. As the MAP decreases within this range, the cerebral vessels dilate to keep CBF constant. As the MAP increases within this range, the cerebral vessels constrict to maintain a constant rate of flow. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 577.*
How does hydrogen ion concentration affect neuronal activity and cerebral blood flow?	An increase in hydrogen ion concentration depresses neuronal activity. It also causes an increase in blood flow to the brain. By doing so, it will help 'wash away' the hydrogen ions, carbon-dioxide, and other acid precursors away from the brain which returns the hydrogen ion concentration back to normal. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 744.*

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Most body tissues can survive for several minutes (and some as long as 30 minutes) without oxygen. Why is the brain not capable of tolerating a similar loss of oxygen?	Non-neural tissues can obtain energy from anaerobic mechanisms, enabling them to acquire energy from glucose and glycogen without the use of oxygen. The brain has a metabolic rate that is about 7.5 times that of non-neural tissues and cannot utilize anaerobic mechanisms effectively. As a result, the brain depends upon second-by-second delivery of oxygen for it to function correctly. Whereas peripheral tissues can tolerate a loss of arterial flow for a considerable period of time before severe damage occurs, the interruption of blood flow to the brain for even 5 seconds can produce unconsciousness. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 749.*
How is PaO2 related to cerebral blood flow?	PaO2 has little effect on CBF until it reaches a tension of 50 mmHg at which point it dramatically increases CBF. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 578.*
What is the normal tissue PO2 in the brain? How do changes in tissue PO2 affect cerebral blood flow?	The normal tissue PO2 in the brain is between 35-40 mmHg. Derangements in cerebral function occur at a tissue PO2 of 20 mmHg. If the tissue PO2 drops below 30 mmHg, the body increases cerebral blood flow to protect the tissue from ischemia. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 744.*
What level can cerebral blood flow fall to before brain damage occurs?	Cerebral blood flow between 15 and 20 mL/100g/min is associated with a flat EEG and values < 10 mL/100g/min are associated with irreversible brain damage. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 576.*
How does hyperglycemia affect the brain during periods of hypoxia?	Hyperglycemia can exacerbate global hypoxic brain injury, therefore glucose containing IV solutions are contraindicated in hypoxic injuries unless a concomitant diagnosis of hypoglycemia has been established. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 576.*
Which of the Rexed laminae are found in the dorsal horn of the spinal column? The ventral horn?	Rexed laminae I through laminae VI are located in the dorsal horn of the spinal column. Laminae VII, VIII, and IX comprise the ventral horn. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 690.*

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Which cranial nerves are sensory only? Which are motor only? Which have both sensory and motor function?

The olfactory, optic, and vestibulocochlear nerves are sensory only. The oculomotor, trochlear, abducens, spinal accessory, and hypoglossal nerves are motor only. The trigeminal, facial, glossopharyngeal, and vagus are both sensory and motor.

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 690.

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